Evaluation Methodology
Analysis and error estimation

Plan

1. **Error estimation techniques**
2. Model comparison and selection

Loss function

- Let f be a model learned on a dataset $S = \{(\mathbf{x}_1, t_1), \ldots, (\mathbf{x}_\ell, t_\ell)\}$. Given a test example x, f returns a prediction $y = f(\mathbf{x})$.
- The **loss function** $\mathcal{L}(t, y)$ measures the cost associated with predicting y instead of t.
- Regression loss functions:
  - **Quadratic loss** : $\mathcal{L}_{quad}(t, y) = (t - y)^2$
  - **Absolute loss** : $\mathcal{L}_{abs}(t, y) = |t - y|$
- Classification loss functions:
  - **0/1 loss** : $\mathcal{L}_{01}(t, y) = 1$ if $y \neq t$, 0 otherwise
    The mean of the 0/1 loss $\equiv$ to the error

Real risk and empirical risk I

- **Risk** (real or expected) Given a model f, a loss function $\mathcal{L}$, and a joint density distribution $p(\mathbf{x}, t)$:

$$\begin{aligned} R[f] &= \int_{\mathbf{x}\times t} \mathcal{L}(t, f(\mathbf{x}))dp(\mathbf{x}, t) \\ &= \int_t \int_{\mathbf{x}} \mathcal{L}(t, f(\mathbf{x}))p(\mathbf{x}, t)d\mathbf{x}dt \end{aligned}$$

Objective : minimise $R[f]$. But since $p(\mathbf{x}, t)$ is unknown, we cannot measure $R[f]$.

Real risk and empirical risk II

- **Empirical risk** given a model f, a loss function $\mathcal{L}$, and a learning dataset S of size ℓ:

$$R_\ell[f] = \frac{1}{\ell} \sum_{i=1}^{\ell} \mathcal{L}(t_i, f(\mathbf{x}_i))$$

However, minimising the empirical risk $\not\Rightarrow$ minimizing the real risk (overfitting).

Error estimation techniques

- **Resubstitution** : estimate the error on the training set
  - underestimates the real error (optimistic bias)
  - favors overfitting on the learning data
- **Hold-out** : use an independent test set. Examples of error estimation scenarios:
  - model evaluation

  - model selection and evaluation

Inconvinience : not all data are used for training

Resampling techniques

- Why resample?
  - Not enough data to do an efficient hold-out
  - the hold-out partition is random. To control randomness we should average over a number of hold-outs
- Resampling techniques
  - Cross-validation
  - Bootstrap

Cross validation

- Divide the data S on V partitions of equal size
- Loop learning/test for V iterations. On each iteration i
  - Keep a partition P_i for test
  - train on $S - P_i$
- Example iteration 5 for $V = 10$

trn	**trn**	**trn**	**trn**	**test**	**trn**	**trn**	**trn**	**trn**	**trn**	**trn**

- Every example is used in learning $V - 1$ times, and in testing 1 time
- Estimate the error on the test data on each iteration and then average over the V iterations

Cross validation : the algorithm

XVAL (S, V)

Divide the dataset S on V equal parts

For $i = 1$ to V do

- $TST_i \leftarrow$ the i^{th}partition
- $TRN_i \leftarrow S - TST_i$
- $h_i \leftarrow$ learn on TRN_i
- $err_i \leftarrow h_i(TST_i)$

Estimated error $\leftarrow$ the mean of the V err_i

Variant of cross validation

- **Stratified cross validation** : each partition retains the original distribution of the target variable
  - Classification : same class distribution
  - Regression : same mean of the target variable
- **Leave-one-out** : cross validation with $V = N = |S|$. On each of the N iterations, we train on $N - 1$ instances and we test on the single example we left out.
- What value for V ?
  - Large V (e.g. leave-one-out) : inreased variance, expensive. To keep for small S
  - small V : smaller variance, but smaller number of samples
  - Compromise the mose oftern used : 10 blocks.

Bootstrap : basic idea

- To estimate the error often we have a single sample S from some population Ω which follows an unknown probability law p
- the estimated error is often optimistically biased
- To correct this bias:
  1. estimate the error $\tilde{\epsilon}$ on S (underestimates the true ϵ)
  2. estimate the bias $\tilde{b}$ by bootstrap resampling
  3. add $\tilde{b}$ to $\tilde{\epsilon}$ in order to have a better error estimation
- How to estimate this optimistic bias?

Simple bootstrap

1. Data : sample S of size N
2. Train a model on S and estimate the error $\tilde{\epsilon}$ on S.
3. For i = 1 to M
   - create a sample S_i by sampling N instances from S with replacement
   - train a model on S_i
   - estimate the error $\tilde{\epsilon}_i$ of the model on the data $S - S_i$
   - estimate the bias $\tilde{b}_i \leftarrow \tilde{\epsilon}_i - \tilde{\epsilon}$
4. the everage bias $\tilde{b} \leftarrow \frac{1}{M} \sum_{i=1}^{M} \tilde{b}_i$
5. The corrected error $\leftarrow \tilde{\epsilon} + \tilde{b}$

In general we need M to be large enough to have a good estimation : between 50 - 200

The bootstrap .632

- When sampling S_i with replacement, where $|S_i| = |S| = N$, an instance may be selected 0 or more times in S_i.
  - Pr($\mathbf{x}$ does not appear in S_i) =

$$(1 - \frac{1}{N})^N$$

  - Pr($\mathbf{x}$ appears at least once in S_i=)

$$1 - (1 - \frac{1}{N})^N \approx 1 - \exp(-1) = 0.632$$

- Modification of the bootstrap algorithm

$$\begin{aligned} \tilde{b}_i &= 0.632(\tilde{\epsilon}_i - \tilde{\epsilon}) \\ \tilde{\epsilon}_{corr_i} &= 0.632(\tilde{\epsilon}_i - \tilde{\epsilon}) + \tilde{\epsilon} \\ &= 0.632\tilde{\epsilon}_i + 0.368\tilde{\epsilon} \end{aligned}$$

- The corrected error on S is the mean of the $\tilde{\epsilon}_{corr_i}$

Plan

1. Error estimation techniques
2. **Model comparison and selection**
   1. **Significance tests**
   2. Performance measures

Model comparison and selection

- "No free lunch theorems" [Wolpert]: no learning algorithm is universally superior of all others.
- The best algorithm is always a function of the problem and the data
- Experimenting is the only way: evaluate different algorithm/parameters for a given problem and choose the best performing model according to the chosen criterion
- Problem of model comparison: some of the differences might be simply random. How to detect the real differences between two models?

The problem

- Let A and B be two learned models evaluated according some performance measure m (e.g. error). We observer that $m_A > m_B$, suggesting that A is better than B
- Two possibilities
  - There is a real difference between the two models $\rightarrow$ choose A
  - The observed difference comes from random variations of the data $\rightarrow$ choose the simplest of the two
- How can we know?
  $\rightarrow$ hypothesis and significance tests

General procedure of model comparison

The quantity that interests us is $\delta = m_A - m_B$ the observed value of which is d

1. Set a null hypothesis H_0 and an alternative hypothesis H_1
   - H_0: $\delta = 0$ (the 2 models are equivalent)
   - H_1: $\delta \neq 0$ (bilateral test), $\delta > 0$, $\delta < 0$ (unilateral test)
2. Fixe α = the desired significance level (e.g. 0.05, 0.01)
3. Estimate the p-value p = the observed significance level
   - significance level = the probability to obtain a δ value as extreme as d if H_0 was real
   - In general we estimate p by calculating the value of the T-test from the observed values of δ
   - The details of this step determine the different test types
4. If $p < \alpha$, reject H_0, otherwise keep H_0

Significance level and test power

	H_0 true	H_0 false
Keep H_0	OK	Type II error
Reject H_0	Type I Error	OK

- α (and p) = probability of a Type I error = P(reject $H_0|H_0$ true). $p < \alpha$: we can reject H_0 because this probability is smaller than the desired minimum
- β = probability of a type II error = P(keep $H_0|H_0$ false)
- $1 - \beta$ = **power** of a test= capacity to distinguish the significant differences: P(reject $H_0|H_0$ false)

Significance tests of the difference of models

- Parametric tests: assume a specific distribution
  e.g. Student's test for part samples (paired t-test).
  - Assume a normal distribution
  - Problem: high risk of Type I error.
- Non parametric tests: they do not assume any distribution
  e.g. McNemar's test

McNemar's test I

1. Set $H_0 : Err_A = Err_B$. $H_1 : Err_A < Err_B$
2. Generate the cross table of errors of A and B

	00 # d´ ex. mal classés par A et B	**01** # d´ex. mal classés par A et bien classés par B	f_{o_1}
f_{o_2}	**10** # d´ex. bien classés par A et mal classés par B	**11** # d´ex. bien classés par A et B	

Only the observed differences f_{o_1} and f_{o_2} are useful (we are interested on performance differences).
According to H_0, the theoretical frequencies f_{t_1} et f_{t_2} are equal:

	00	***(01+10)/2***	f_{t_1}
f_{t_2}	***(01+10)/2***	*11*	

McNemar's test II

3. Calculate the value of the χ^2 test : quantify the deviation of the observed frequencies from the theoretical frequencies

$$
\begin{aligned}
\chi^2 &= \sum_{i=1}^{2} \frac{(f_{o_i} - f_{t_i})^2}{f_{t_i}} \\
&= \frac{(01 - (01+10)/2)^2}{(01+10)/2} + \frac{(10 - (01+10)/2)^2}{(01+10)/2} \\
&= \frac{(01-10)^2}{01+10} \\
&= \frac{(|01-10|-1)^2}{01+10} \text{ after the Yates correction}
\end{aligned}
$$

4. Calculate p the significance level of the observed χ^2 value with $df = 1$. If $p < \alpha$ reject H_0 otherwise keep it.

Yates's correction

- The statistics χ^2 : is calculated from discrete data (frequences)
- to find the p-value p, we use the continuous distribution χ^2
- Result: p underestimated, higher risk of a type I error
- To reduce that risk we adjust the χ^2 statistics downwards, since for fixed df $p \nearrow$ when $\chi^2 \searrow$
- Not a clear consensus among experts on when to use this correction. According to some useful only for 2×2 tables.

Multiple comparisons and Bonferroni adjustements

- A hypothesis test is applied on two models. To find a good model we often do $J > 2$ tests
- The result of the multiplicity $\rightarrow$ higher risk of Type I error
- Bonferroni adjustment
  - Calculate the number of tests J
    m models against a reference model : $J = m$
    m models against each other: $J = \begin{pmatrix} m \\ 2 \end{pmatrix}$ tests
  - For J tests, replace the significance threshold α by α/J
  - Reject H_0 if $p < \alpha/J$

  Inverse danger: increase the risk of a type II error

Plan

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   1. Significance tests
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Coincidence matrices

Visualise the repartition between correct and wrong classification

	cible		
préd.	C_1	...	C_k
P_1			
...			
P_k			

- M_{ii} number of correct classifications for class i
- M_{ij} $i \neq j$: number of instances of the j class classified as i

$$ACC = \frac{1}{N} \sum_{i=1}^{N} M_{ii}$$

- Accuracy
- Error $\epsilon = 1 - ACC$

Performance measures

- Confusion matrix in binary classification

	Target		
Pred.	c_+	c_-	
p_+	**TP**	FP	$\lvert p_+\rvert$
p_-	FN	**TN**	$\lvert p_-\rvert$
	$\lvert c_+\rvert$	$\lvert c_-\rvert$	N

- Sensitivity=recall=TPR (true positive rate)$= P(p_+|c_+) = \frac{TP}{|c_+|}$
- Specificity=TNR (true negative rate) $= P(p_-|c_-) = \frac{TN}{|c_-|}$
- FPR (false positive rate) $= 1-$ specificity $= P(p_+|c_-) = \frac{FP}{|c_-|}$
- Precision (rate of correct positive predictions) $= P(c_+|p_+) = \frac{TP}{|p_+|}$

Limitations of accuracy / error rate

- Imbalanced classes : accuracy is dominated by the majority class
  - The geometric mean of the accuracies avoids that

$$MG_{ACC} = [\prod_{i=1}^{k} \frac{1}{|c_i|} M_{ii}]^{1/k}$$

  - In binary classification, MG_{ACC} =sensitivity * specificity
- Ofter errors do not have the same cost
  e.g. false negatives can be deadly in medicine

Misclassification cost

- Create a table of costs to weight the error rate

Err	c_+	c_-
p_+	TP	**FP**
p_-	**FN**	TN

Cost	c_+	c_-
p_+	0	**c(FP)**
p_-	**c(FN)**	0

Ex. Churn

- c(FP) = price of an offer given to retain a client
- c(FN) = loss of a client

- Total missclassification cost

$$Cout_{mis} = FP * c(FP) + FN * c(FN)$$